

Student Guide — primary / module-14- magic-squares

IKS Curriculum

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Student Workbook

Student Workbook — Magic Squares — Bhadragaṇita (Primary)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *bhadra-gaṇita* — _____
- *siddhāṅka* (modern coinage) — _____
- *kothumi vidhi* — _____

Day-by-day notes

Day 1 — Discover the 3×3

What I learned:

One question I have:

Day 2 — The Magic Constant

What I learned:

One question I have:

Day 3 — How Many 3×3 Magic Squares?

What I learned:

One question I have:

Day 4 — Stair-Step Method (Nārāyaṇa)

What I learned:

One question I have:

Day 5 — Practice — 5×5 and 7×7

What I learned:

One question I have:

Day 6 — 4×4 and the Khajuraho Square

What I learned:

One question I have:

Day 7 — Historical Context — Nārāyaṇa and Predecessors

What I learned:

One question I have:

Day 8 — Cross-Cultural — Lo Shu and Dürer

What I learned:

One question I have:

Day 9 — Design Your Own

What I learned:

One question I have:

Day 10 — Final Share and Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — Discover the 3×3

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Build the unique 3×3 magic square with 1–9 by trial.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. “auspicious arithmetic” — the Indian term for magic squares”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Discover the 3×3**. Keep sentences short.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 2 — The Magic Constant

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Derive why 3×3 with 1–9 must sum to 15 per row.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *siddhāṅka* (modern coinage) (IAST: siddhāṅka (modern coinage); lit. “the common sum of all rows/columns/diagonals”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **The Magic Constant**. Keep sentences short.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 3 — How Many 3×3 Magic Squares?

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Count the equivalence classes (up to symmetry).

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa's algorithm for odd-order squares”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **How Many 3×3 Magic Squares?**. Keep sentences short.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 4 — Stair-Step Method (Nārāyaṇa)

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Apply Nārāyaṇa's algorithm to build a 5×5 square.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. “auspicious arithmetic” — the Indian term for magic squares)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Stair-Step Method (Nārāyaṇa)**. Keep sentences short.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 5 — Practice — 5×5 and 7×7

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Use the method on multiple odd-order squares.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *siddhāṅka* (modern coinage) (IAST: siddhāṅka (modern coinage); lit. “the common sum of all rows/columns/diagonals”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Practice** — 5×5 and 7×7. Keep sentences short.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
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- If energy is low, swap the activity for free drawing.

Day 6 — 4×4 and the Khajuraho Square

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Examine the Khajuraho inscription and its panel-magic property.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa's algorithm for odd-order squares”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **4×4 and the Khajuraho Square**. Keep sentences short.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 7 — Historical Context — Nārāyaṇa and Predecessors

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Place *Gaṇita-kaumudī* in 14th c. Indian math.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Historical Context — Nārāyaṇa and Predecessors**. Keep sentences short.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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- If energy is low, swap the activity for free drawing.

Day 8 — Cross-Cultural — Lo Shu and Dürer

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Compare Indian, Chinese (Lo Shu), and European (Dürer 1514) magic squares.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *siddhāṅka* (modern coinage) (IAST: siddhāṅka (modern coinage); lit. “the common sum of all rows/columns/diagonals”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Cross-Cultural — Lo Shu and Dürer**. Keep sentences short.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 9 — Design Your Own

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Create a personalised 4×4 or 5×5 magic square as visual art.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa's algorithm for odd-order squares”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Design Your Own**. Keep sentences short.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Final Share and Assessment

Module: Magic Squares — Bhadragaṇita (Primary) · **Time:** 30 min

Goal

Present design + summative quiz.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Final Share and Assessment**. Keep sentences short.

Activity (8 min)

A simplified version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Homework

Homework — Magic Squares — Bhadragaṇita (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — Discover the 3×3** · Draw about today's concept. (~5 min)
2. **Day 2 — The Magic Constant** · Draw about today's concept. (~5 min)
3. **Day 3 — How Many 3×3 Magic Squares?** · Draw about today's concept. (~5 min)
4. **Day 4 — Stair-Step Method (Nārāyaṇa)** · Draw about today's concept. (~5 min)
5. **Day 5 — Practice — 5×5 and 7×7** · Draw about today's concept. (~5 min)
6. **Day 6 — 4×4 and the Khajuraho Square** · Draw about today's concept. (~5 min)
7. **Day 7 — Historical Context — Nārāyaṇa and Predecessors** · Draw about today's concept. (~5 min)
8. **Day 8 — Cross-Cultural — Lo Shu and Dürer** · Draw about today's concept. (~5 min)
9. **Day 9 — Design Your Own** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Share and Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.